Midterm Exam — Answer Key

FinTech 545 — Quantitative Risk Management

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About This Key

Every number below comes from `solution.jl`, which uses the functions in `library/`. Problem 3(c) is simulated. For each simulated value we report the median of 1,000 runs of 5,000 draws, followed by the 99% band, the 0.5% and 99.5% quantiles across those runs. A student answer from a single run of 5,000 draws should fall inside the band. Building the copula correlation from Spearman correlation instead of Kendall's τ also lands inside every band.

1 Pairwise Correlation with Missing Data

The 50 rows hold only 18 on which all five series are observed. Each pair is jointly observed on 28 to 36 days, so every pairwise entry is estimated on a different sample.

1.1 Part (a)

Table 1: Pairwise correlation

	x1	x2	x3	x4	x5
x1	1.0000	0.3328	0.4185	0.2553	0.7323
x2	0.3328	1.0000	0.2510	0.2877	0.7875
x3	0.4185	0.2510	1.0000	0.4836	0.6843
x4	0.2553	0.2877	0.4836	1.0000	0.7826
x5	0.7323	0.7875	0.6843	0.7826	1.0000

1.2 Part (b)

Each variance uses every observation of that series, 36 to 42 days.

Table 2: Full sample variance

	x1	x2	x3	x4	x5
Variance $\times 10^4$	4.0607	4.8007	1.5111	8.4174	2.7315
Standard deviation	0.02015	0.02191	0.01229	0.02901	0.01653

1.3 Part (c)

The matrix is not PSD. Its eigenvalues are:

$$-0.0893, \quad 0.4613, \quad 0.7191, \quad 0.8234, \quad 3.0854$$

A PSD matrix has no negative eigenvalue. The eigenvector of the negative eigenvalue is $(-0.287, -0.365, -0.157, -0.342, 0.802)$. That is a portfolio long `x5` and short the other four, and the pairwise matrix assigns it a negative variance.

The cause is the estimation, not the data. `x5` is highly correlated with all four of the others, so the true matrix has a small eigenvalue in that direction. Each correlation was estimated on a different set of days, and nothing forces the ten estimates to be consistent with each other. On roughly 30 days each, sampling noise is easily large enough to push the small eigenvalue below zero. The complete case matrix, all on the same 18 rows, is PSD with a smallest eigenvalue of 0.0123.

1.4 Part (d)

We apply Higham's method to the correlation matrix with $W = I$. It converges in 26 iterations.

Table 3: Nearest PSD correlation, Higham

	x1	x2	x3	x4	x5
x1	1.0000	0.3495	0.4259	0.2708	0.6946
x2	0.3495	1.0000	0.2606	0.3075	0.7393
x3	0.4259	0.2606	1.0000	0.4925	0.6626
x4	0.2708	0.3075	0.4925	1.0000	0.7377
x5	0.6946	0.7393	0.6626	0.7377	1.0000

The eigenvalues are now:

$$0.0000, \quad 0.4501, \quad 0.7035, \quad 0.8067, \quad 3.0397$$

The negative eigenvalue moves to zero and the others shrink slightly. The Frobenius distance from the pairwise matrix is 0.1214.

Table 4: Change in each entry, Higham minus pairwise

	x1	x2	x3	x4	x5
x1	0	0.0166	0.0075	0.0155	-0.0377
x2	0.0166	0	0.0096	0.0198	-0.0482
x3	0.0075	0.0096	0	0.0089	-0.0217
x4	0.0155	0.0198	0.0089	0	-0.0450
x5	-0.0377	-0.0482	-0.0217	-0.0450	0

The `x5` correlations moved the most, each down by 0.02 to 0.05. The other correlations rose by up to 0.02. This lines up with the eigenvector from part (c), which loads most heavily on `x5`. Higham moves the matrix the shortest distance that removes the negative direction, and that direction is mostly `x5` against the rest.

1.5 Part (e)

The covariance is $\Sigma = DRD$, where R is the Higham correlation from part (d) and D is the diagonal matrix of full sample standard deviations from part (b).

Table 5: Covariance

$\times 10^4$	x1	x2	x3	x4	x5
x1	4.0607	1.5429	1.0551	1.5831	2.3132
x2	1.5429	4.8007	0.7019	1.9549	2.6773
x3	1.0551	0.7019	1.5111	1.7565	1.3461
x4	1.5831	1.9549	1.7565	8.4174	3.5372
x5	2.3132	2.6773	1.3461	3.5372	2.7315

Scaling by D cannot create a negative eigenvalue, so Σ is PSD. Its smallest eigenvalue is zero to machine precision, so it is singular rather than positive definite. A simulation from it needs a Cholesky that tolerates PSD input, or the PCA approach.

2 VaR and ES for a Single Stock

Grading note: Remove 5 points from part (a) and 5 points from part (b) if the student fits the OLS regression to estimate the distribution of `XYZ`. Check the code for each part, not only the reported numbers. A student who fits the model and rebuilds the variance from $\widehat{\beta}$ and the residual variance lands within 0.002 percentage points of the correct VaR (see the check at the end of this problem), so the numbers alone will not show it. Deduct nothing if the student fits the regression but states that it is not necessary.

2.1 Method

The regression is not needed. Under the model, the return of `XYZ` is a linear combination of two normal variables, so it is itself normal:

$$r_{XYZ} \sim N\left(\alpha + \beta\mu_{SPY},\ \beta^2\sigma^2_{SPY} + \sigma^2_\epsilon\right)$$

OLS fit on the same sample gives back the sample mean and the sample variance of `XYZ`. The fitted line passes through the means, and the variance splits exactly into the explained and the residual parts. Everything the model can tell us about the distribution of `XYZ` is already in the mean and variance of its own returns.

From the 500 arithmetic returns of `XYZ`:

- Mean: 3.8×10^{-18}, zero to machine precision.
- Standard deviation: $\sigma = 0.019604$.

With $\mu = 0$:

$$\text{VaR}_\alpha = -\Phi^{-1}(\alpha)\,\sigma \qquad \text{ES}_\alpha = \sigma\,\frac{\phi\left(\Phi^{-1}(\alpha)\right)}{\alpha}$$

2.2 Part (a)

Table 6: VaR of XYZ

	Multiplier	VaR
95%	1.6449	3.2245%
99%	2.3263	4.5605%

2.3 Part (b)

Table 7: ES of XYZ

	Multiplier	ES
95%	2.0627	4.0436%
99%	2.6652	5.2248%

On the 100 shares, worth \$7,451.83 at the last price, that is a 95% VaR of \$240.28 and a 95% ES of \$301.33.

As a check, fitting the model gives $\hat{\beta} = 1.1933$ and a residual standard deviation of 0.014605. Rebuilding the variance from the pieces gives a standard deviation of 0.019614, which is 1.00056 times the sample value. The gap is the $n-2$ in the residual variance against the $n-1$ in the sample variance. It moves the 95% VaR from 3.2245% to 3.2263%. The model adds work and a little noise, and nothing else.

3 Copula VaR and ES for a Two Book Portfolio

3.1 Part (a)

Table 8: Sample moments of the arithmetic returns

	Mean	SD	Skew	Excess Kurtosis
S1	0.000904	0.011826	-0.040	-0.033
S2	0.000800	0.015348	0.071	5.477
S3	0.000424	0.018898	-1.321	4.696
S4	0.000780	0.009933	-0.035	-0.133
S5	0.000709	0.018951	-0.307	2.733
S6	0.000805	0.015085	-1.202	4.700
S7	0.000693	0.013674	-0.041	-0.123
S8	0.000374	0.017112	-0.444	6.529
S9	0.001746	0.020647	-1.083	4.461
S10	0.000165	0.013061	-0.696	5.583

The moments sort the stocks before any fitting. `S1`, `S4`, and `S7` show no skew and no excess kurtosis. `S3`, `S6`, and `S9` are skewed left by more than one. The other four have fat tails with modest skew. We fit all three families to every stock by maximum likelihood, without removing the mean.

Table 9: Normal and t fits

	Normal μ	Normal σ	t μ	t σ	t ν
S1	0.000904	0.011826	0.000904	0.011819	15357
S2	0.000800	0.015348	0.000581	0.011673	4.858
S3	0.000424	0.018898	0.002291	0.013657	4.161
S4	0.000780	0.009933	0.000780	0.009927	9935
S5	0.000709	0.018951	0.000833	0.015577	6.289
S6	0.000805	0.015085	0.002445	0.010655	3.824
S7	0.000693	0.013674	0.000693	0.013666	15588
S8	0.000374	0.017112	0.000654	0.011424	3.483
S9	0.001746	0.020647	0.003383	0.015310	4.385
S10	0.000165	0.013061	0.000405	0.009427	4.121

Table 10: NIG fits

	NIG μ	NIG α	NIG β	NIG δ
S1	0.076557	2086.49	-588.20	0.257473
S2	-0.000305	73.21	4.86	0.016593
S3	0.010474	79.84	-36.35	0.019657
S4	0.089549	3368.19	-984.96	0.290287
S5	0.001497	73.58	-2.23	0.025958
S6	0.008272	91.68	-41.69	0.014628
S7	0.077482	1635.96	-443.71	0.272611
S8	0.001183	50.62	-2.89	0.014161
S9	0.010128	63.16	-22.93	0.021515

	NIG μ	NIG α	NIG β	NIG δ
S10	0.001049	76.69	-5.37	0.012595

We choose with AICc and check the choice with BIC. Each column below is the gap to the best model under that criterion, so the winner shows 0.

Table 11: Model selection, gap to the best fit

	ΔAICc Normal	ΔAICc t	ΔAICc NIG	ΔBIC Normal	ΔBIC t	ΔBIC NIG	Choice
S1	0	2.01	3.78	0	6.91	13.57	Normal
S2	114.65	0	5.30	109.76	0	10.19	t
S3	185.13	43.18	0	175.34	38.29	0	NIG
S4	0	2.02	3.85	0	6.91	13.64	Normal
S5	59.97	0	4.66	55.08	0	9.55	t
S6	190.19	48.66	0	180.41	43.76	0	NIG
S7	0	2.02	3.81	0	6.91	13.59	Normal
S8	198.18	0	6.19	193.29	0	11.09	t
S9	146.00	28.75	0	136.22	23.86	0	NIG
S10	139.53	0	4.16	134.64	0	9.05	t

AICc and BIC agree on every stock:

- **Normal** – `S1`, `S4`, `S7`. The t fit drives ν past 9,000 and the NIG drives α into the thousands. Both have collapsed onto the normal. Their log likelihoods are within 0.13 of the normal's, and the extra parameters cost more than that. At α in the thousands the NIG density underflows the Bessel function in a direct evaluation, so we computed its log density with the exponentially scaled Bessel function.
- t – `S2`, `S5`, `S8`, `S10`. Fitted ν runs from 3.5 to 6.3. The NIG is second each time, 4 to 6 points behind on AICc. The normal is far behind.
- **NIG** – `S3`, `S6`, `S9`. Fitted β is clearly negative, matching the left skew. The symmetric t trails by 29 to 49 points on AICc because it cannot bend to the skew.

The normal against the t can never be a large AICc win. The t nests the normal as $\nu \to \infty$, so the most the normal can win by is the penalty for one parameter, about 2.

3.2 Part (b)

We transform each stock to uniforms through the margin chosen in part (a), then estimate the correlation matrix from Kendall's τ, with $\rho = \sin(\pi\tau/2)$. Both copulas use the same R, so the comparison isolates ν. The Gaussian copula adds no parameters beyond R, and the t adds one.

Table 12: Copula correlation from Kendall's τ

	S1	S2	S3	S4	S5	S6	S7	S8	S9	S10
S1	1.00	0.54	0.53	0.56	0.54	0.29	0.25	0.23	0.28	0.29
S2	0.54	1.00	0.56	0.53	0.52	0.33	0.23	0.25	0.31	0.28
S3	0.53	0.56	1.00	0.53	0.57	0.32	0.30	0.27	0.34	0.30
S4	0.56	0.53	0.53	1.00	0.57	0.31	0.23	0.27	0.30	0.30
S5	0.54	0.52	0.57	0.57	1.00	0.33	0.27	0.28	0.28	0.30
S6	0.29	0.33	0.32	0.31	0.33	1.00	0.54	0.52	0.54	0.55
S7	0.25	0.23	0.30	0.23	0.27	0.54	1.00	0.53	0.52	0.54
S8	0.23	0.25	0.27	0.27	0.28	0.52	0.53	1.00	0.52	0.52
S9	0.28	0.31	0.34	0.30	0.28	0.54	0.52	0.52	1.00	0.52
S10	0.29	0.28	0.30	0.30	0.30	0.55	0.54	0.52	0.52	1.00

The matrix has a clear block structure: 0.52 to 0.57 inside each book and 0.23 to 0.34 across books. It is PD, with a smallest eigenvalue of 0.382.

Table 13: Copula fits

	Log Likelihood	k	$\hat{\nu}$	AICc	BIC
Gaussian	2110.02	0	–	-4220.05	-4220.05
t	2108.16	1	100	-4214.32	-4209.42

The Gaussian copula fits better. It wins by 5.72 on AICc and 10.63 on BIC.

The fitted ν sits on the upper bound of the search, 100. The t copula log likelihood is still rising there: 1756.0 at $\nu = 5$, 1997.0 at 10, 2074.3 at 20, 2103.0 at 50, and 2108.2 at 100. The data want $\nu \to \infty$, which is the Gaussian copula. The fat tails in this portfolio live in the margins, not in the dependence between the stocks.

3.3 Part (c)

We simulate the Gaussian copula with the R from part (b) and push each column of uniforms through its fitted margin from part (a). The simulated returns keep their fitted means, so the VaR and ES are absolute, not relative to the mean. Each position is the holding times the last price. P&L is the simulated value less the current value, summed to each book and to the total.

Table 14: VaR and ES at 95%, in dollars

Position	Value	VaR 95%	99% Band	ES 95%	99% Band
S1	48,329	898	[855, 947]	1,134	[1,084, 1,187]
S2	46,697	1,078	[1,014, 1,150]	1,563	[1,450, 1,690]
S3	24,267	804	[747, 867]	1,202	[1,123, 1,302]
S4	43,597	678	[648, 712]	858	[820, 897]
S5	37,727	1,103	[1,039, 1,170]	1,540	[1,445, 1,648]
S6	46,327	1,213	[1,122, 1,307]	1,850	[1,725, 1,991]

Position	Value	VaR 95%	99% Band	ES 95%	99% Band
S7	40,144	875	[835, 919]	1,105	[1,055, 1,154]
S8	26,046	645	[602, 692]	1,016	[926, 1,131]
S9	98,249	3,379	[3,125, 3,629]	5,110	[4,731, 5,512]
S10	24,512	479	[450, 512]	718	[654, 788]
Book A	200,618	3,622	[3,434, 3,854]	4,895	[4,633, 5,157]
Book B	235,278	5,428	[4,996, 5,829]	7,834	[7,267, 8,378]
Total	435,896	7,664	[7,177, 8,104]	10,648	[9,932, 11,349]

Grading note: The discussion is open. Give full credit for any discussion that shows the student understood what they calculated. The paragraphs below are one example, not a list of required points.

Diversification shows up at every level. The stock VaRs in book A sum to \$4,560 against a book VaR of \$3,622. In book B they sum to \$6,591 against \$5,428. The two book VaRs sum to \$9,049 against a total of \$7,664. ES tells the same story: \$12,729 across the two books against \$10,648 for the total.

Book B carries more risk per dollar than book A, 2.31% of value at 95% VaR against 1.81%. `S9` is the largest position, a NIG margin with a heavy left tail. Its ES of \$5,110 is more than half of the \$9,799 sum of stand-alone ES in book B.